

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.013058	0.013324	27.997
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143

Forward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.002130	0.002586	4.378
harmonic_regression_fw	wave1_current_registry	Fw	0.062598	0.062702	133.783
periodic_mlp_fw	wave1_current_registry	Fw	0.002118	0.002575	4.351
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.002234	0.002695	4.616
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.038372	0.039223	82.451

Backward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713

Backward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.002680	0.003213	4.759
harmonic_regression_bw	wave1_current_registry	Bw	0.008041	0.008675	16.236
periodic_mlp_bw	wave1_current_registry	Bw	0.002700	0.003230	4.793
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.002713	0.003255	4.822
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.013927	0.014623	28.984

Forward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.002210	0.002686	4.557
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.002078	0.002536	4.254
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.002083	0.002538	4.261
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.002004	0.002385	4.123
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.001195	0.001461	2.559
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.001730	0.002084	3.517

Backward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.002566	0.003101	4.747
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002440	0.002971	4.477
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002430	0.002973	4.452
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.002482	0.002972	4.647
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.001129	0.001412	2.228
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.001405	0.001757	2.700

Forward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.045021	0.045184	96.691
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.037261	0.038590	80.109
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.038030	0.039604	81.958
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.042694	0.042917	91.847
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.034610	0.035965	74.503
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.033574	0.035202	72.338

Backward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010936	0.011364	22.574
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.020530	0.024251	43.464
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.015162	0.018761	32.047
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.011676	0.012103	23.908
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.016792	0.019189	35.128
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.015864	0.020039	33.657

Forward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.036294	0.036467	77.932

Backward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.008945	0.009670	18.730

Forward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.045185	0.045455	97.083
wave3_2_harmonic_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.059869	0.059969	127.873

Backward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.008376	0.009258	17.556
wave3_2_harmonic_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.012996	0.013533	26.826

Forward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.044577	0.044713	95.253
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.037030	0.037237	79.717
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.058349	0.058552	124.991
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.050091	0.050255	107.778

Backward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.009939	0.010532	20.584
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.010075	0.010673	20.657
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.011255	0.011824	23.166
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.012620	0.013521	25.713

Forward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.049061	0.049181	105.141
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.058171	0.058283	124.441
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.054964	0.055089	117.915

Backward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.014320	0.014747	30.084
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.012012	0.012561	24.411
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.009813	0.010421	19.966

Forward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_fw	wave4_2_probabilistic_registry	Fw	0.050281	0.050425	107.720
wave4_2_gaussian_nll_fw	wave4_2_probabilistic_registry	Fw	0.054985	0.055109	117.675

Backward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_bw	wave4_2_probabilistic_registry	Bw	0.011462	0.011942	23.503
wave4_2_gaussian_nll_bw	wave4_2_probabilistic_registry	Bw	0.009739	0.010201	20.349

Forward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_fw	wave4_3_mixture_density_registry	Fw	0.047438	0.047677	102.227
wave4_3_mixture_density_k3_fw	wave4_3_mixture_density_registry	Fw	0.057840	0.057965	124.009

Backward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_bw	wave4_3_mixture_density_registry	Bw	0.008282	0.010642	16.490
wave4_3_mixture_density_k3_bw	wave4_3_mixture_density_registry	Bw	0.011511	0.011910	23.777

Forward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.024419	0.025009	52.319
wave4_4_causal_tcn_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.028476	0.028960	61.107

Backward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.016484	0.016958	34.591
wave4_4_causal_tcn_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.007773	0.008896	15.746

Forward Wave 3 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.046994	0.047257	100.353
wave5_1_harmonic_prior_smooth_l1_structured_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.047938	0.048105	102.360

Backward Wave 3 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.009485	0.010128	19.928
wave5_1_harmonic_prior_smooth_l1_structured_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.011508	0.012082	24.388

Forward Wave52b Offset Harmonic Guided Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52b_offset_centered_shape_harmonic_fw	wave52b_offset_harmonic_guided_registry	Fw	0.001695	0.002045	3.391

Backward Wave52b Offset Harmonic Guided Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52b_offset_centered_shape_harmonic_bw	wave52b_offset_harmonic_guided_registry	Bw	0.002266	0.002708	3.986

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
periodic_mlp_fw	wave1_current_registry	Fw	0.002118	0.002575	4.351
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
feedforward_fw	wave1_current_registry	Fw	0.002130	0.002586	4.378

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
feedforward_bw	wave1_current_registry	Bw	0.002680	0.003213	4.759
periodic_mlp_bw	wave1_current_registry	Bw	0.002700	0.003230	4.793
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.002713	0.003255	4.822

Forward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.002210	0.002686	4.557
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.002078	0.002536	4.254
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.002083	0.002538	4.261
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.002004	0.002385	4.123
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.001195	0.001461	2.559
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.001730	0.002084	3.517

Backward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.002566	0.003101	4.747
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002440	0.002971	4.477
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002430	0.002973	4.452
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.002482	0.002972	4.647
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.001129	0.001412	2.228
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.001405	0.001757	2.700

Forward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.045021	0.045184	96.691
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.037261	0.038590	80.109
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.038030	0.039604	81.958
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.042694	0.042917	91.847
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.034610	0.035965	74.503
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.033574	0.035202	72.338

Backward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010936	0.011364	22.574
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.020530	0.024251	43.464
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.015162	0.018761	32.047
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.011676	0.012103	23.908
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.016792	0.019189	35.128
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.015864	0.020039	33.657

Forward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.036294	0.036467	77.932

Backward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.008945	0.009670	18.730

Forward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.044577	0.044713	95.253
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.037030	0.037237	79.717
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.058349	0.058552	124.991
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.050091	0.050255	107.778

Backward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.009939	0.010532	20.584
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.010075	0.010673	20.657
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.011255	0.011824	23.166
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.012620	0.013521	25.713

Forward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.029204	0.029350	62.562
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.062756	0.062835	134.143
tree_fw	wave1_current_registry	Fw	0.002125	0.002612	4.355
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.049061	0.049181	105.141
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.058171	0.058283	124.441
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.054964	0.055089	117.915

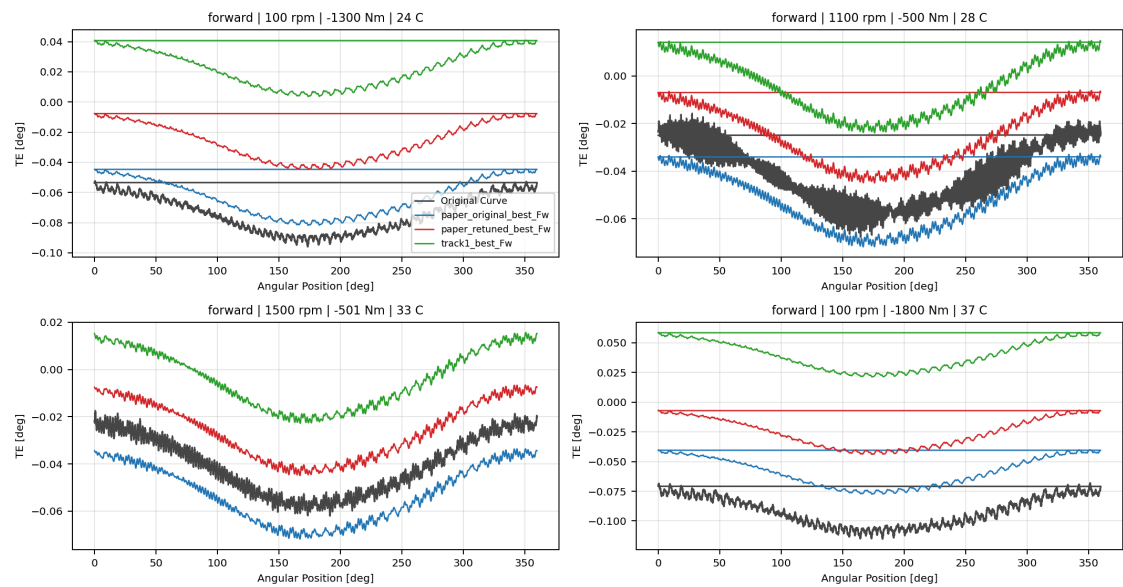
Backward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.005363	0.006139	10.674
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.006633	0.007119	13.713
tree_bw	wave1_current_registry	Bw	0.002756	0.003287	4.934
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.014320	0.014747	30.084
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.012012	0.012561	24.411
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.009813	0.010421	19.966

Comparison Gallery - Forward Reference Model Overlay

Included models: `paper_original_best_Fw` , `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` .

Forward Reference Model Overlay

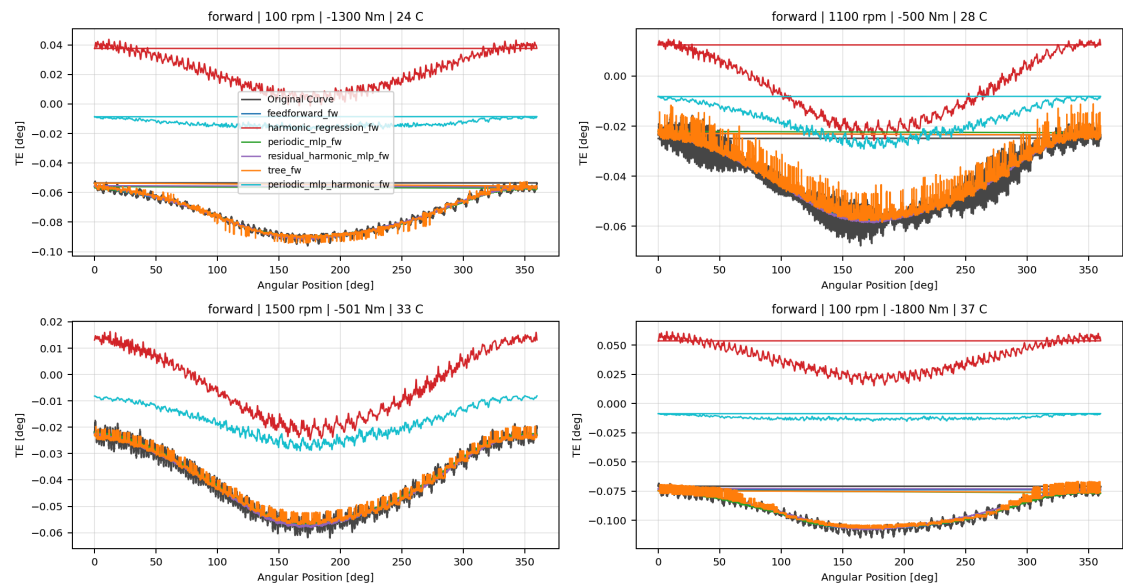


Forward Reference Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 1 Family Model Overlay

Included models: `feedforward_fw` , `harmonic_regression_fw` , `periodic_mlp_fw` , `residual_harmonic_mlp_fw` , `tree_fw` , `periodic_mlp_harmonic_fw` .

Forward Wave 1 Family Model Overlay

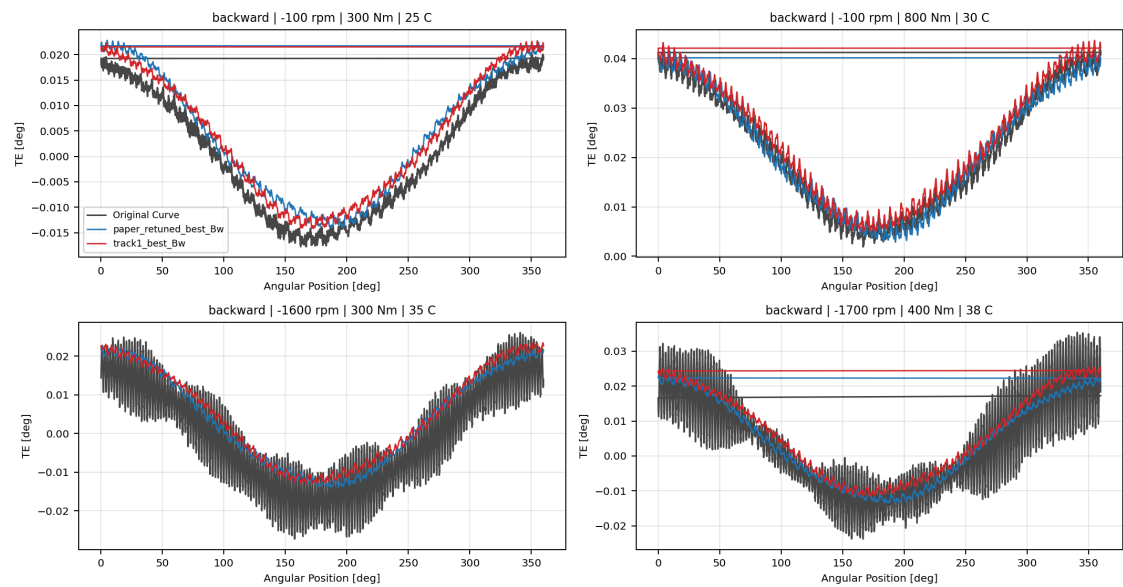


Forward Wave 1 Family Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Model Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` .

Backward Reference Model Overlay

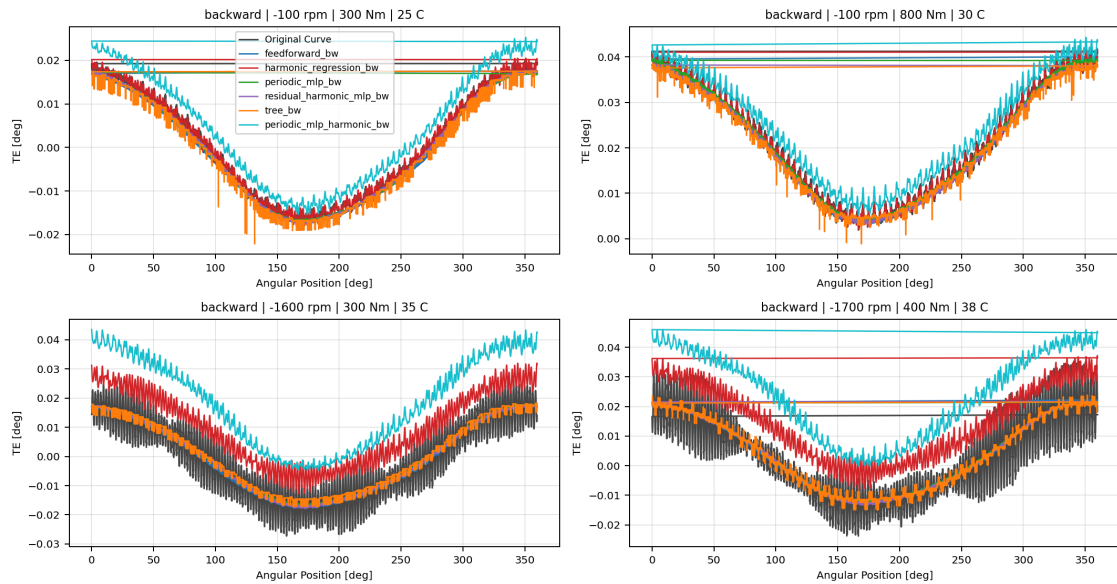


Backward Reference Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 1 Family Model Overlay

Included models: `feedforward_bw` , `harmonic_regression_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` , `tree_bw` , `periodic_mlp_harmonic_bw` .

Backward Wave 1 Family Model Overlay

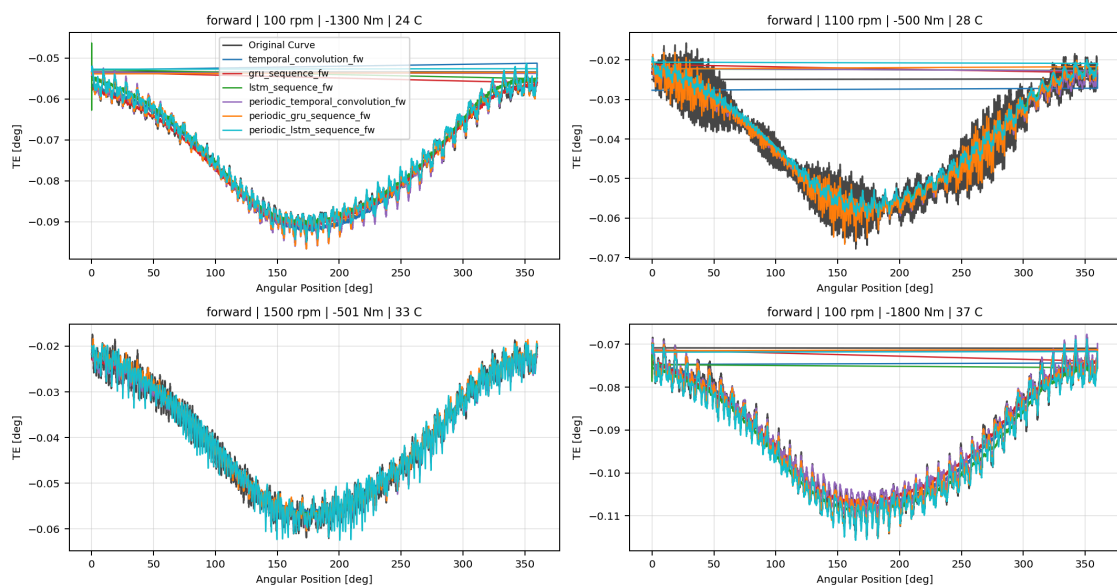


Backward Wave 1 Family Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Wave 2.1 Temporal Model Overlay

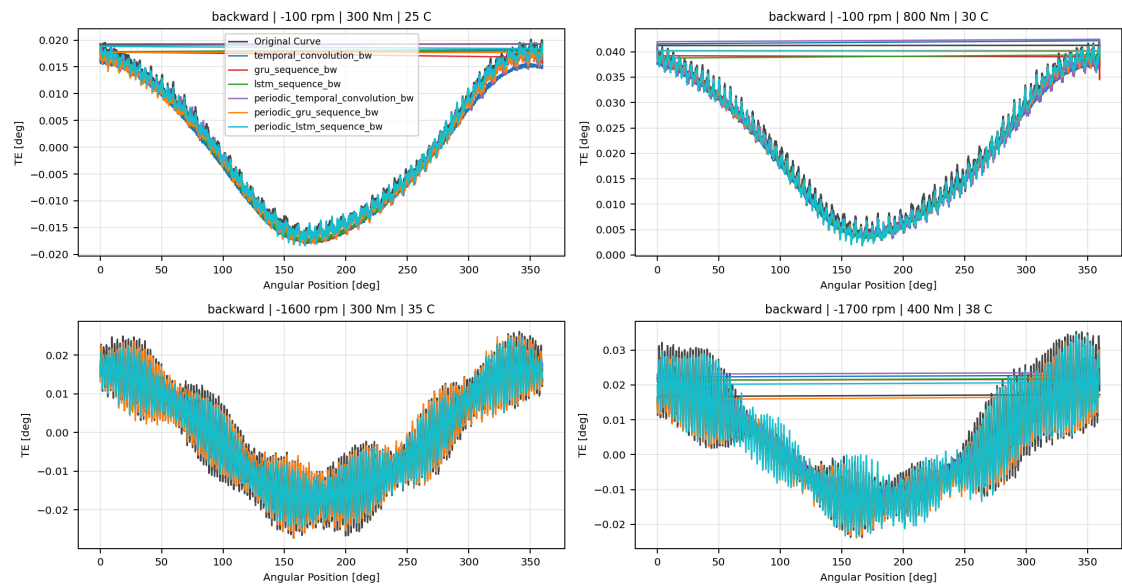


Forward Wave 2.1 Temporal Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Wave 2.1 Temporal Model Overlay

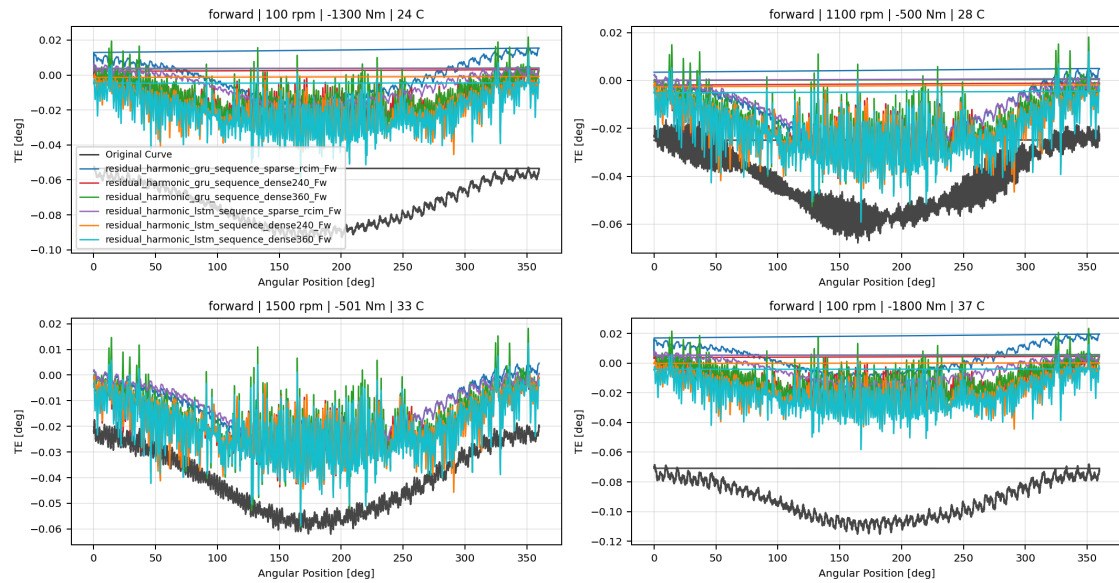


Backward Wave 2.1 Temporal Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Fw` ,
`residual_harmonic_gru_sequence_dense240_Fw` , `residual_harmonic_gru_sequence_dense360_Fw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Fw` , `residual_harmonic_lstm_sequence_dense240_Fw` ,
`residual_harmonic_lstm_sequence_dense360_Fw` .

Forward Wave 2.3 Residual Harmonic Temporal Overlay

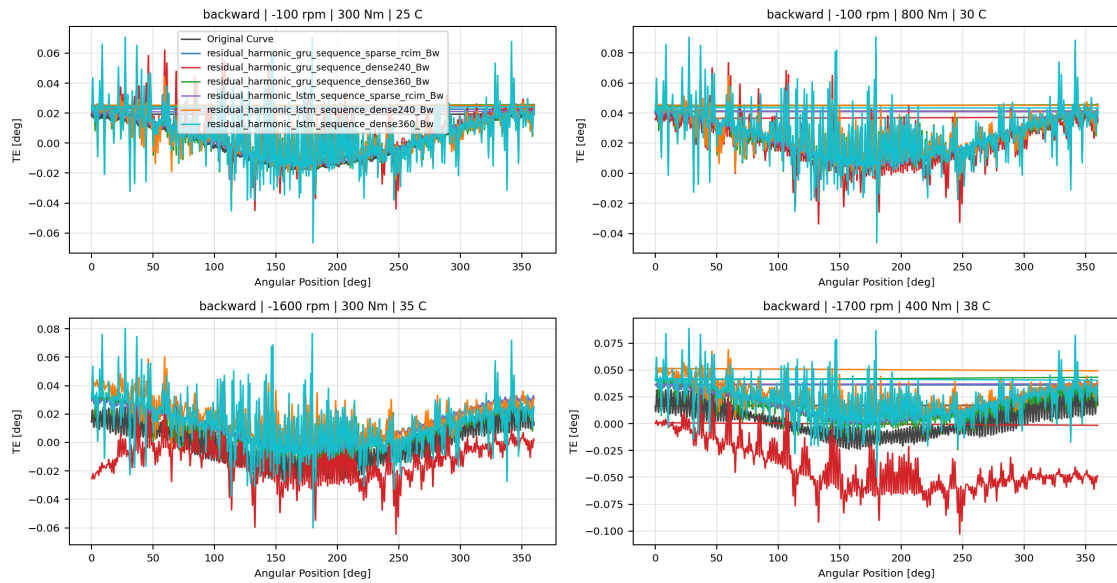


Forward Wave 2.3 Residual Harmonic Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Bw` ,
`residual_harmonic_gru_sequence_dense240_Bw` , `residual_harmonic_gru_sequence_dense360_Bw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Bw` , `residual_harmonic_lstm_sequence_dense240_Bw` ,
`residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Wave 2.3 Residual Harmonic Temporal Overlay

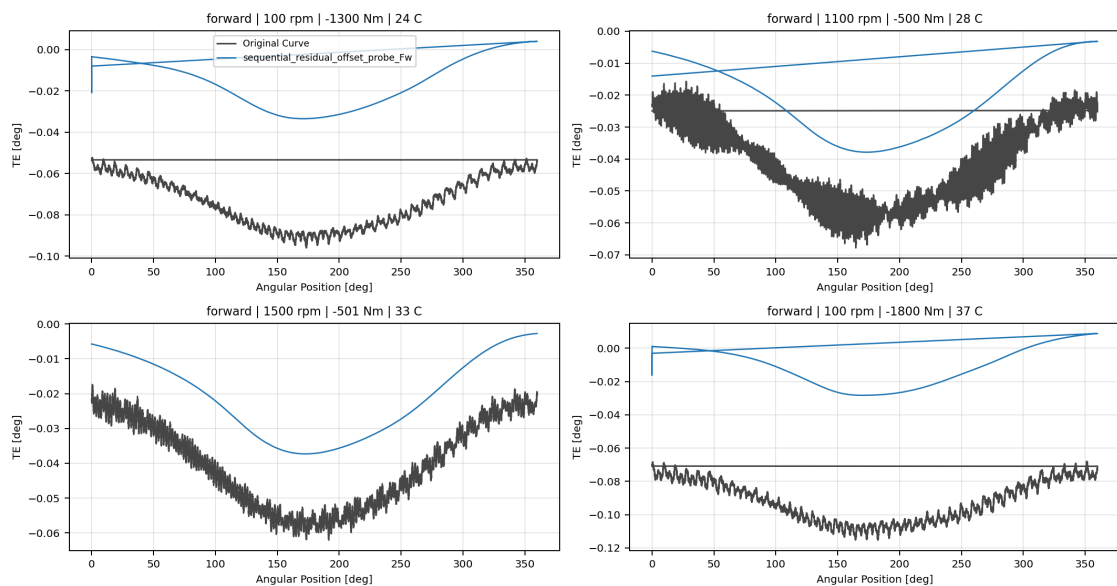


Backward Wave 2.3 Residual Harmonic Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_fw` .

Forward Wave 3.1 Offset-Aware Probe Overlay

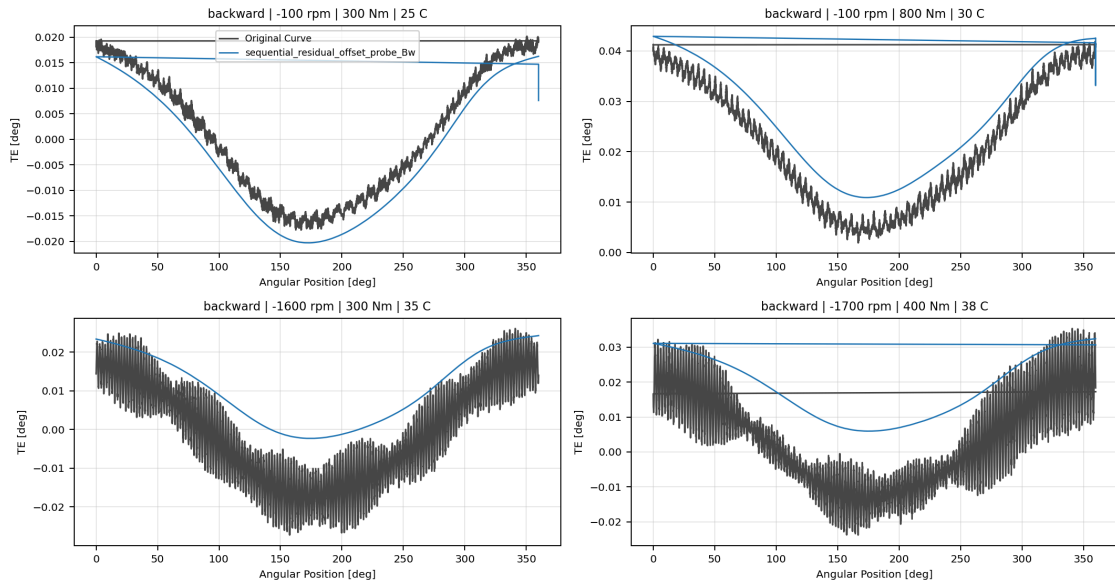


Forward Wave 3.1 Offset-Aware Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_bw` .

Backward Wave 3.1 Offset-Aware Probe Overlay

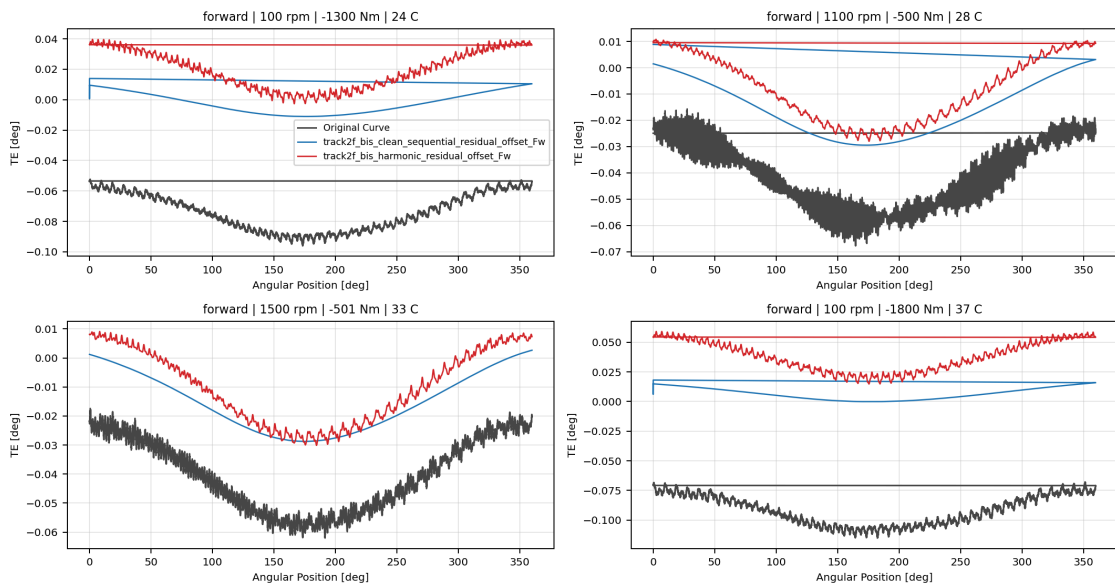


Backward Wave 3.1 Offset-Aware Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_fw` , `wave3_2_harmonic_residual_offset_fw` .

Forward Wave 3.2 Harmonic-Offset Probe Overlay

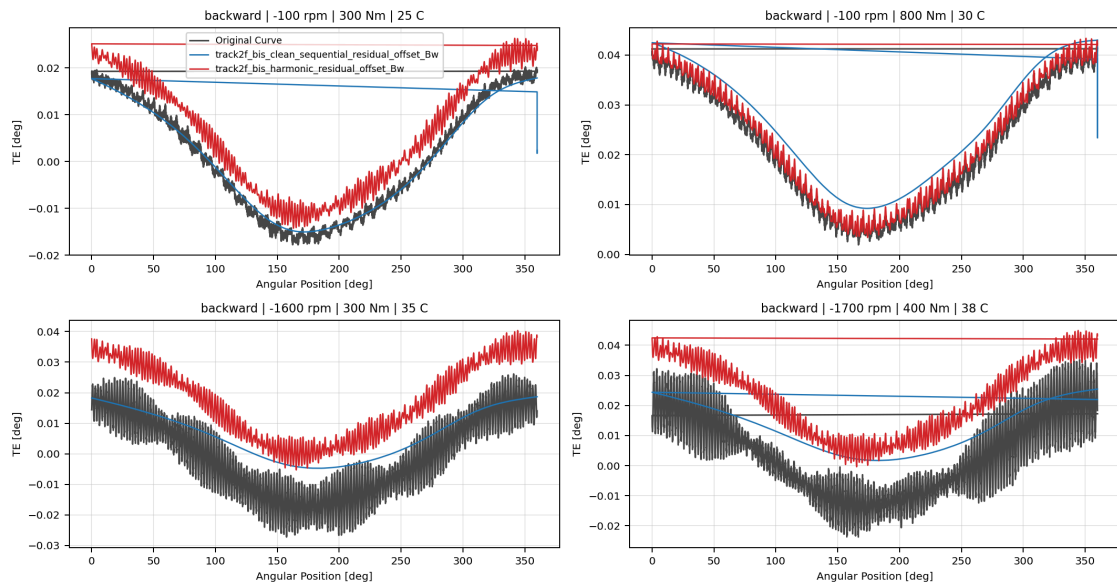


Forward Wave 3.2 Harmonic-Offset Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_bw` , `wave3_2_harmonic_residual_offset_bw` .

Backward Wave 3.2 Harmonic-Offset Probe Overlay

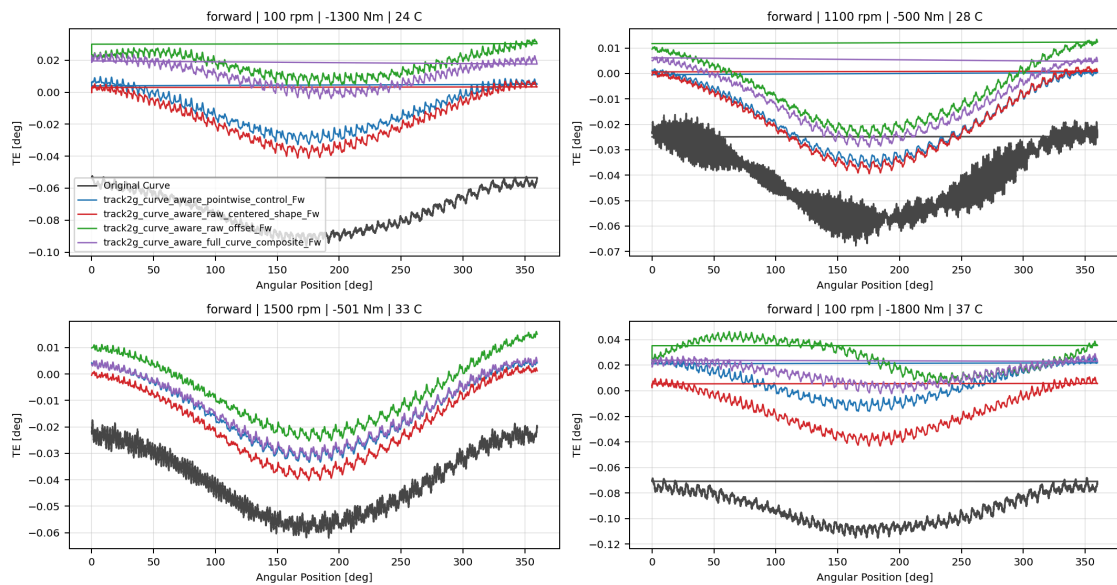


Backward Wave 3.2 Harmonic-Offset Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` , `wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Wave 3.3 Curve-Aware Training Overlay

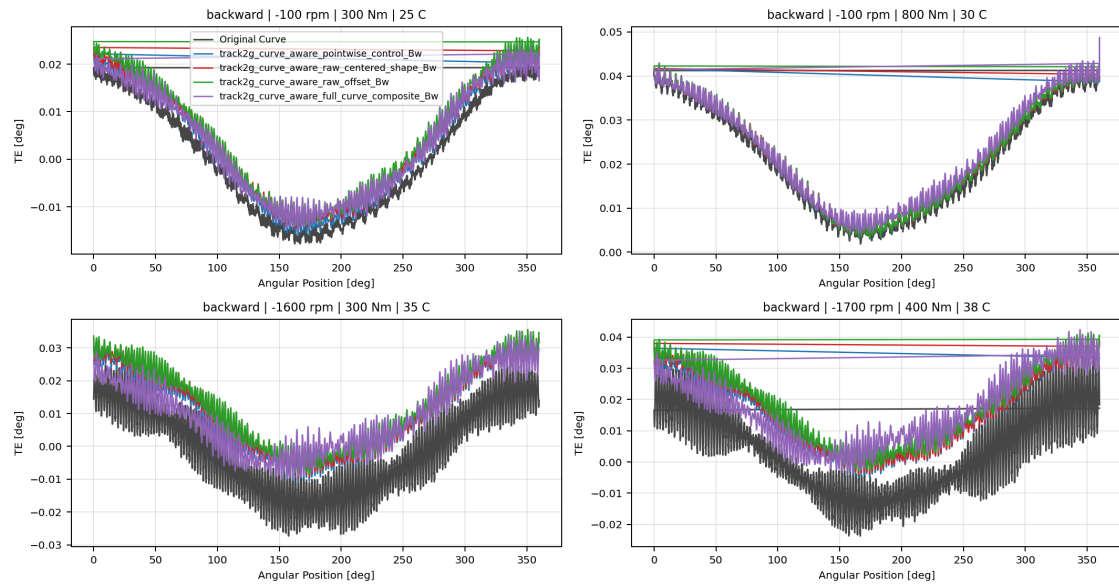


Forward Wave 3.3 Curve-Aware Training Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_bw` , `wave3_3_raw_centered_shape_curve_aware_bw` , `wave3_3_raw_offset_curve_aware_bw` , `wave3_3_full_curve_composite_bw` .

Backward Wave 3.3 Curve-Aware Training Overlay

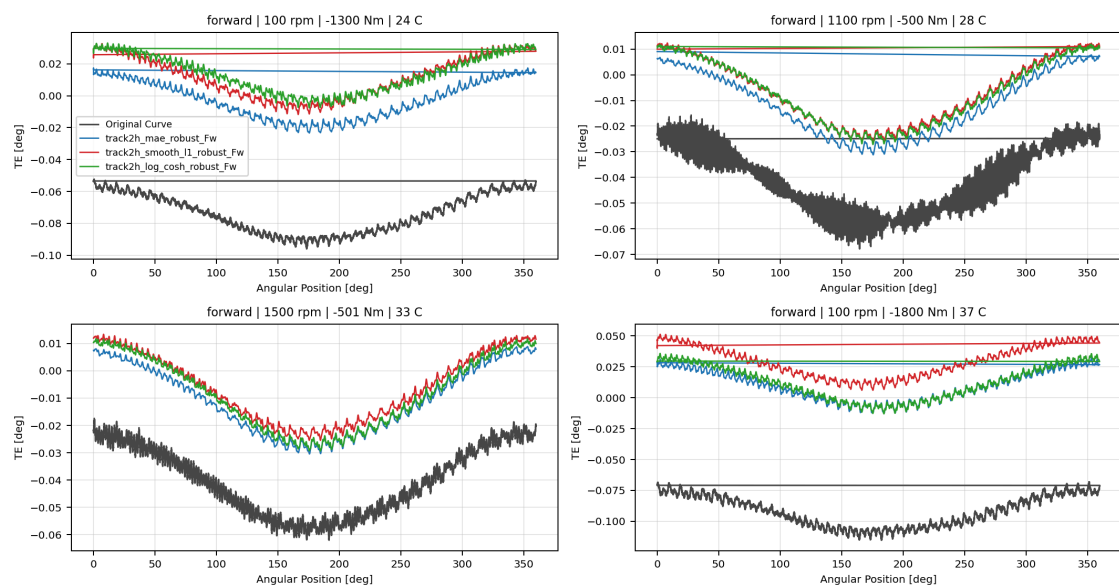


Backward Wave 3.3 Curve-Aware Training Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_fw` , `wave4_1_smooth_l1_robust_loss_fw` , `wave4_1_log_cosh_robust_loss_fw` .

Forward Wave 4.1 Robust-Loss Overlay

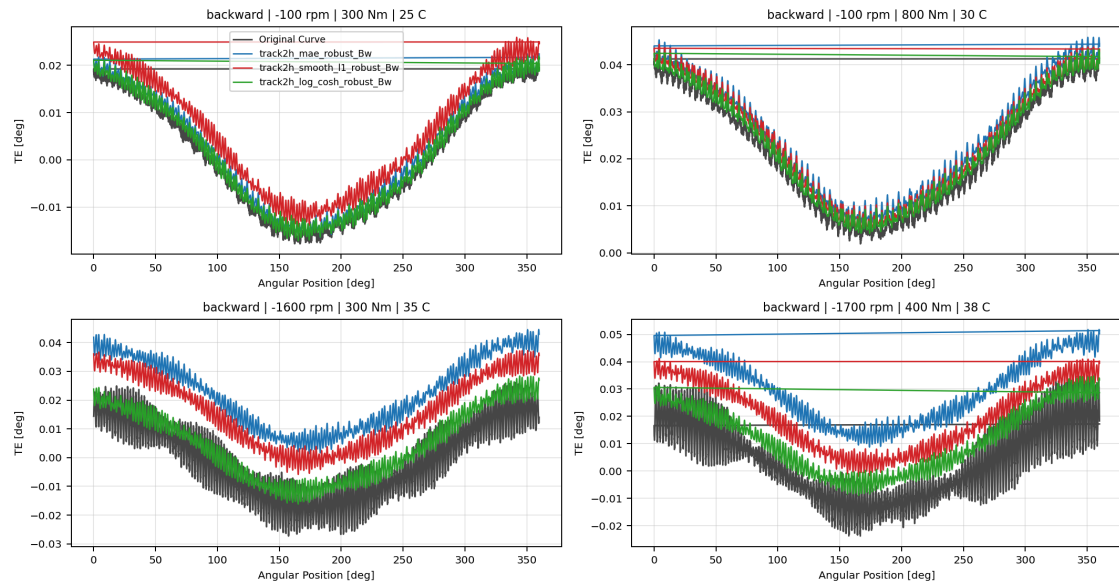


Forward Wave 4.1 Robust-Loss Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_bw` , `wave4_1_smooth_l1_robust_loss_bw` ,
`wave4_1_log_cosh_robust_loss_bw` .

Backward Wave 4.1 Robust-Loss Overlay

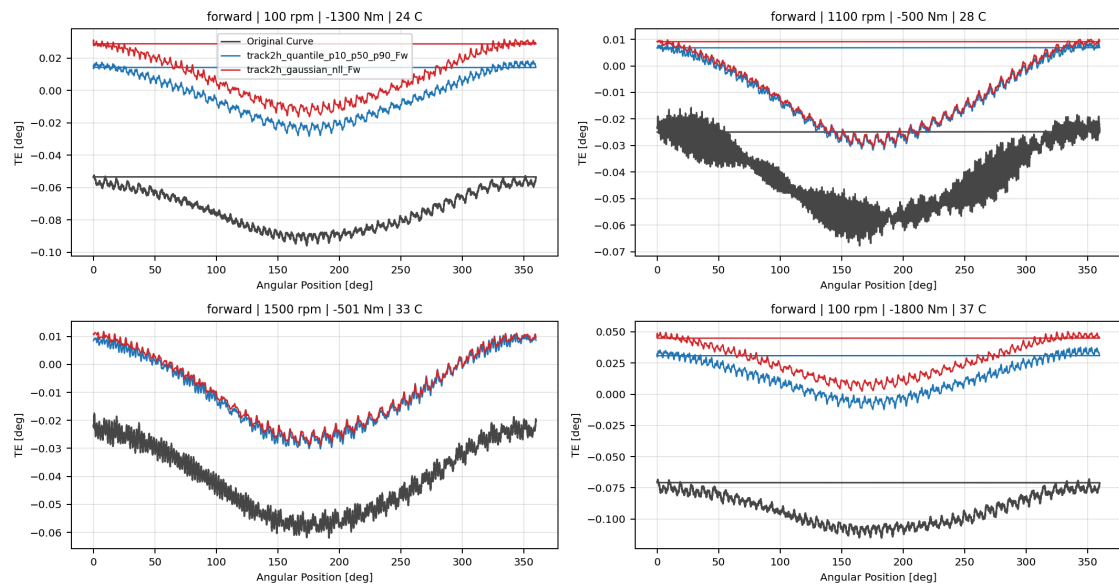


Backward Wave 4.1 Robust-Loss Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_fw` , `wave4_2_gaussian_nll_fw` .

Forward Wave 4.2 Quantile Probabilistic Overlay

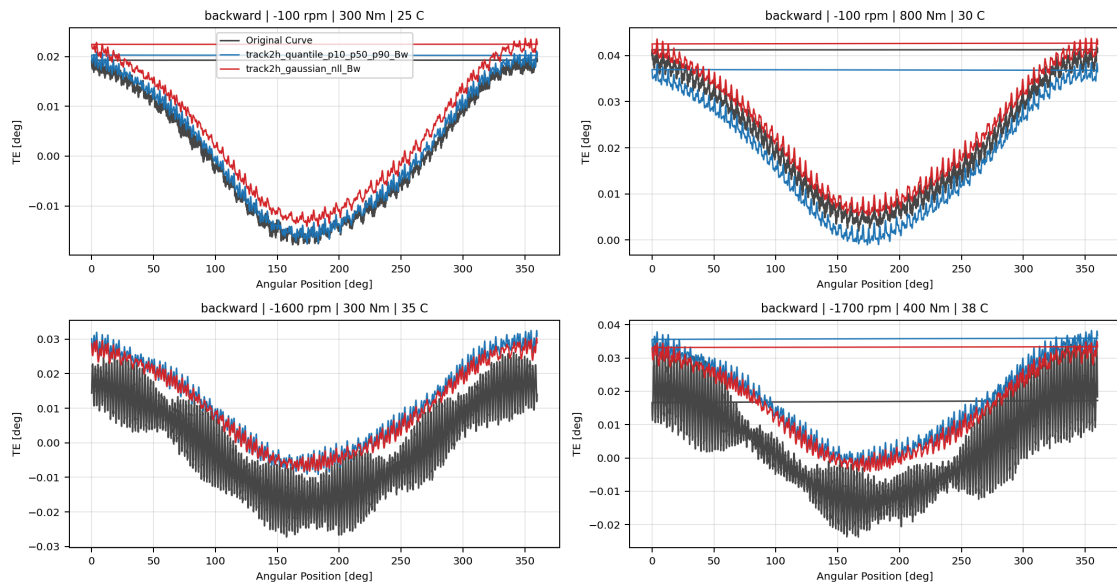


Forward Wave 4.2 Quantile Probabilistic Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_bw` , `wave4_2_gaussian_nll_bw` .

Backward Wave 4.2 Quantile Probabilistic Overlay

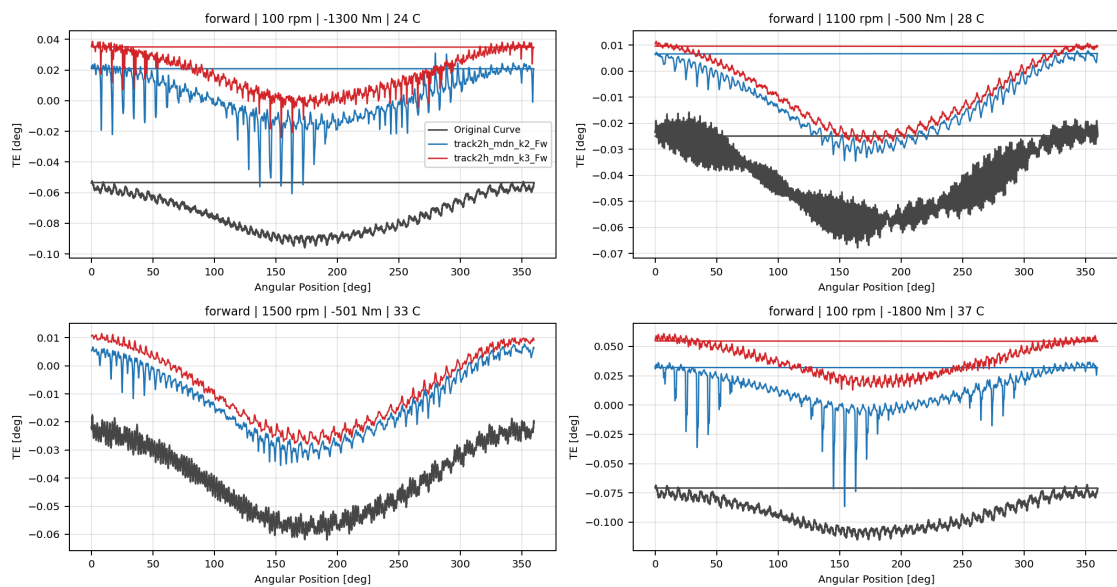


Backward Wave 4.2 Quantile Probabilistic Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_fw` , `wave4_3_mixture_density_k3_fw` .

Forward Track2h Mixture Density Heads Registry Overlay

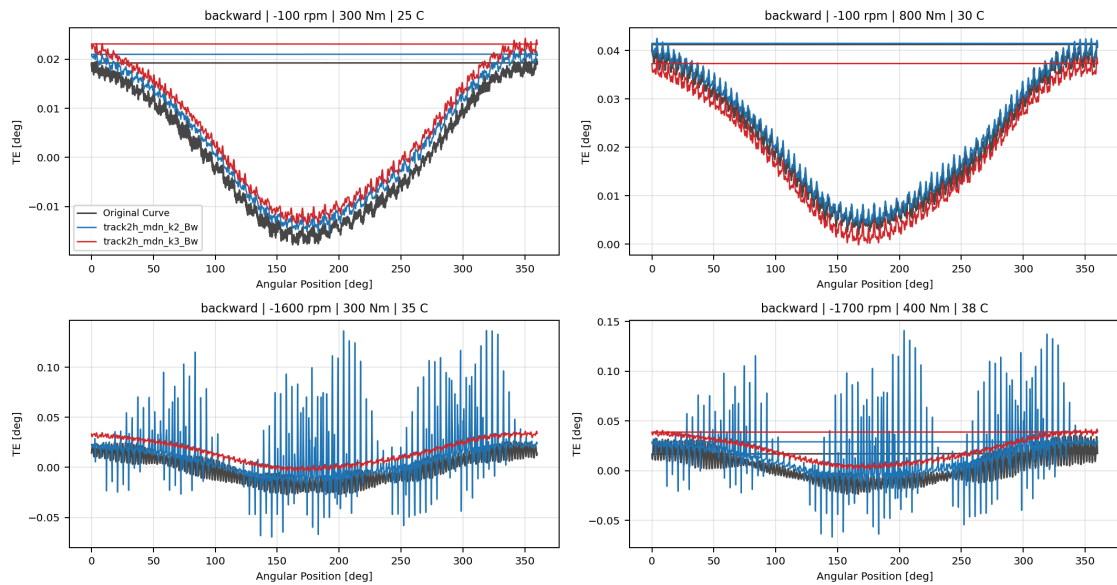


Forward Wave 4.3 Mixture Density Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_bw` , `wave4_3_mixture_density_k3_bw` .

Backward Track2h Mixture Density Heads Registry Overlay

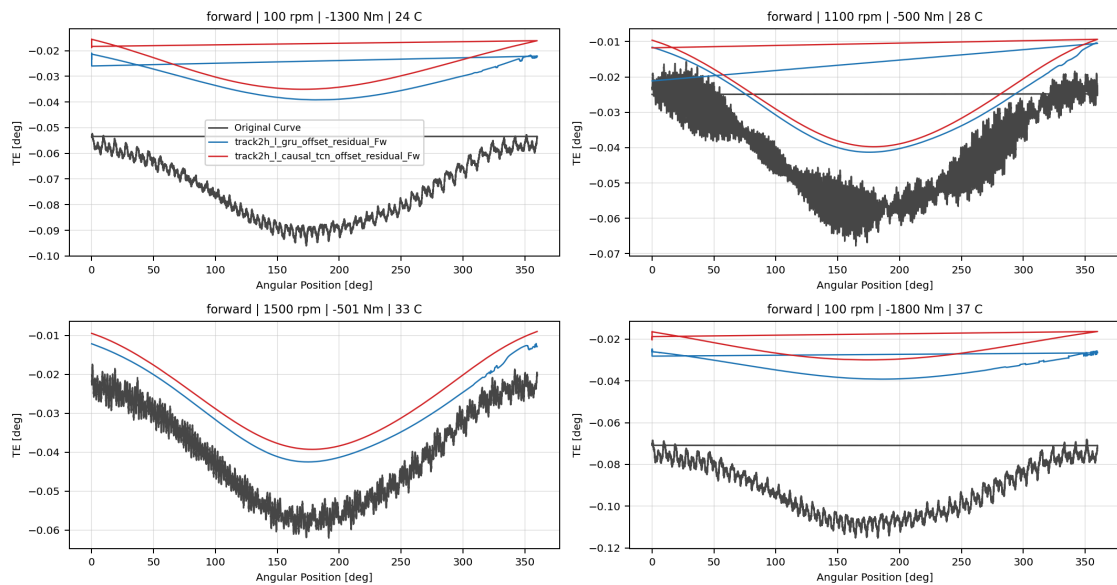


Backward Wave 4.3 Mixture Density Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_fw` , `wave4_4_causal_tcn_latent_offset_residual_fw` .

Forward Track2h Latent State Hysteresis Registry Overlay

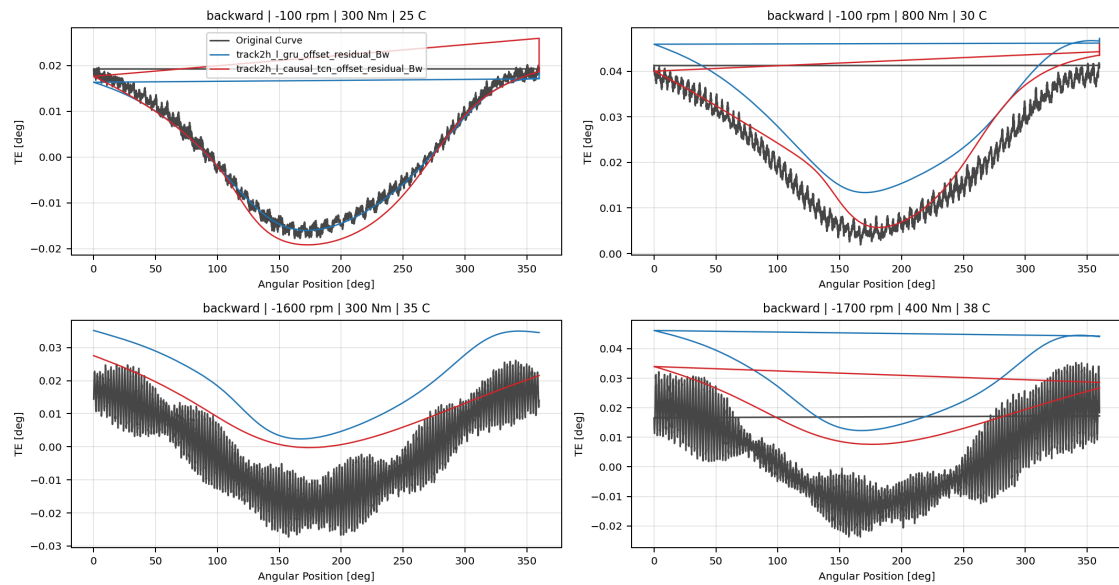


Forward Wave 4.4 Latent State Hysteresis Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_bw` , `wave4_4_causal_tcn_latent_offset_residual_bw` .

Backward Track2h Latent State Hysteresis Registry Overlay

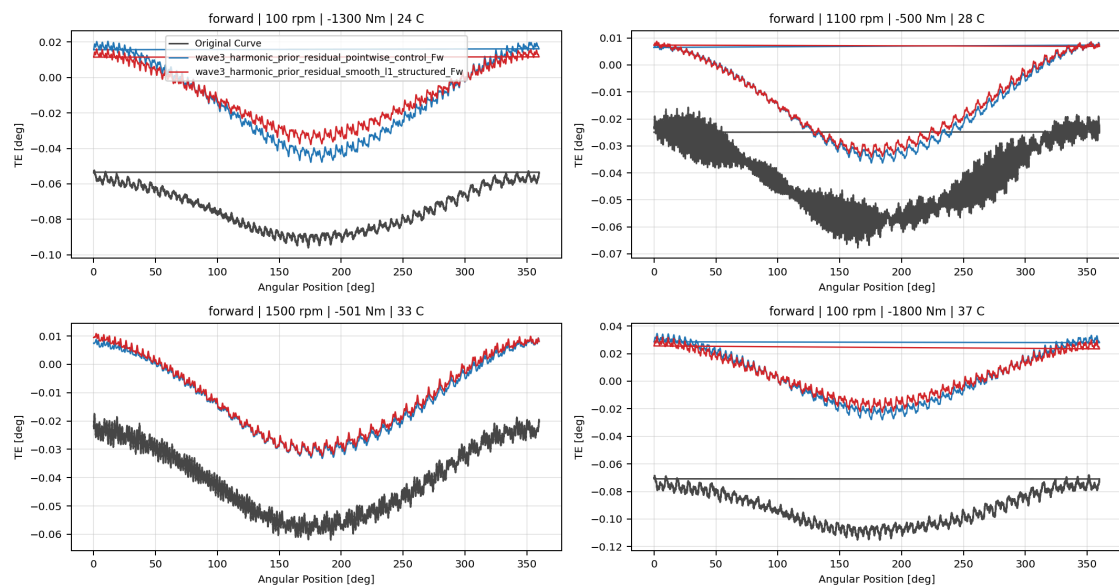


Backward Wave 4.4 Latent State Hysteresis Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3 Harmonic Prior Residual Registry Overlay

Included models: `wave5_1_harmonic_prior_pointwise_control_fw` ,
`wave5_1_harmonic_prior_smooth_l1_structured_fw` .

Forward Wave 3 Harmonic Prior Residual Registry Overlay

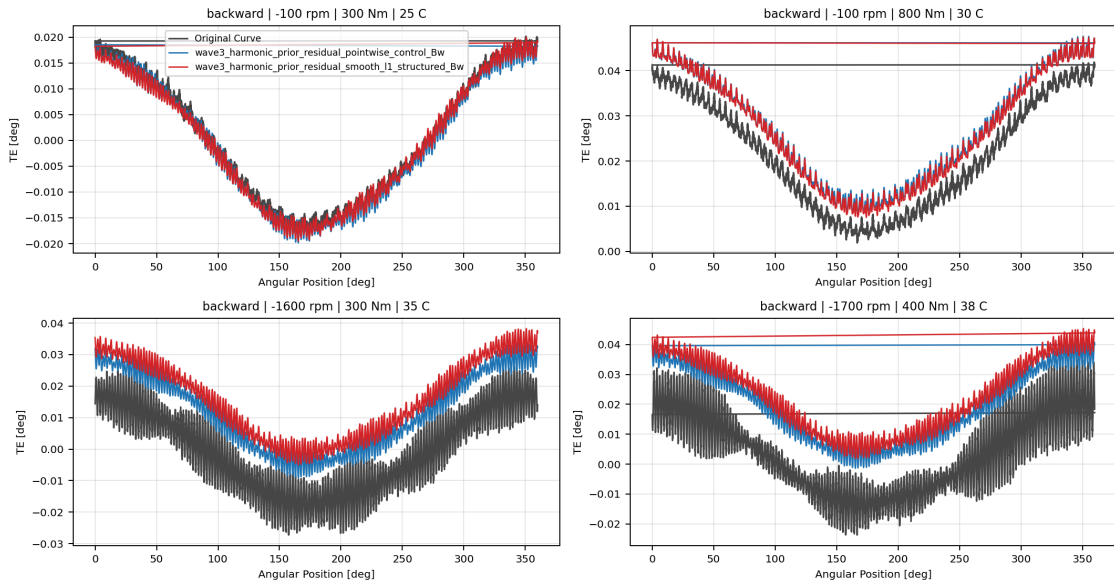


Forward Wave 3 Harmonic Prior Residual Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3 Harmonic Prior Residual Registry Overlay

Included models: `wave5_1_harmonic_prior_pointwise_control_bw` ,
`wave5_1_harmonic_prior_smooth_l1_structured_bw` .

Backward Wave 3 Harmonic Prior Residual Registry Overlay

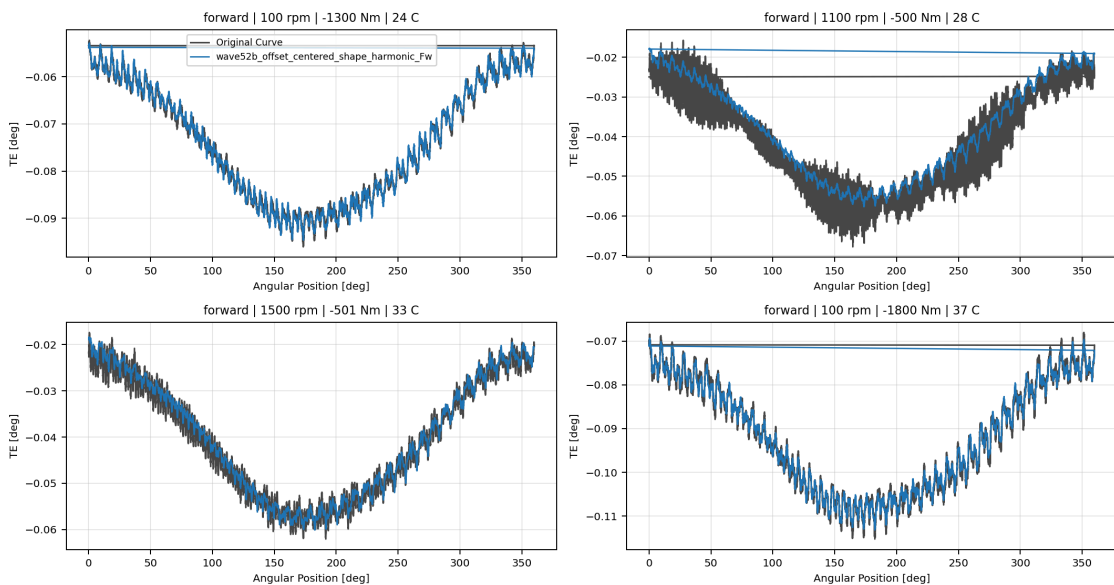


Backward Wave 3 Harmonic Prior Residual Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave52b Offset Harmonic Guided Registry Overlay

Included models: `wave52b_offset_centered_shape_harmonic_Fw` .

Forward Wave52b Offset Harmonic Guided Registry Overlay

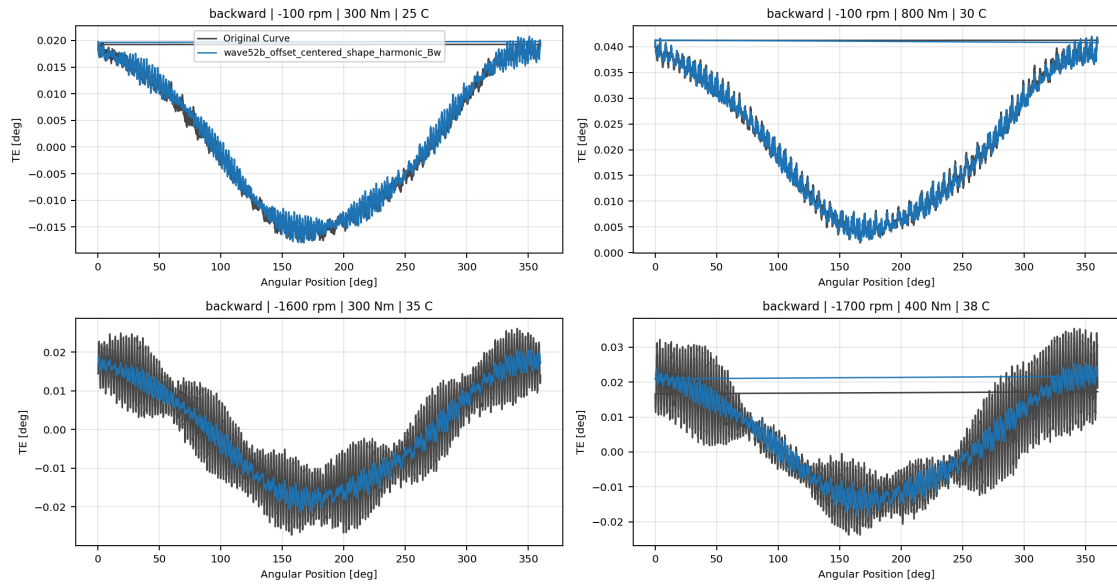


Forward Wave52b Offset Harmonic Guided Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave52b Offset Harmonic Guided Registry Overlay

Included models: `wave52b_offset_centered_shape_harmonic_Bw` .

Backward Wave52b Offset Harmonic Guided Registry Overlay

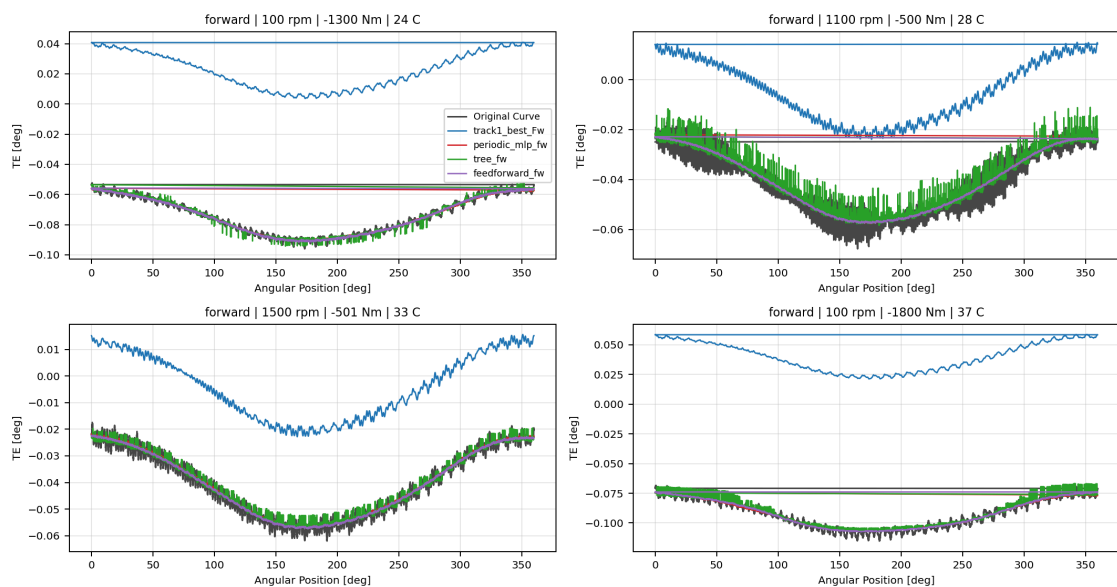


Backward Wave52b Offset Harmonic Guided Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: `rcim_model_bank_reproduction_best_fw` , `periodic_mlp_fw` , `tree_fw` , `feedforward_fw` .

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

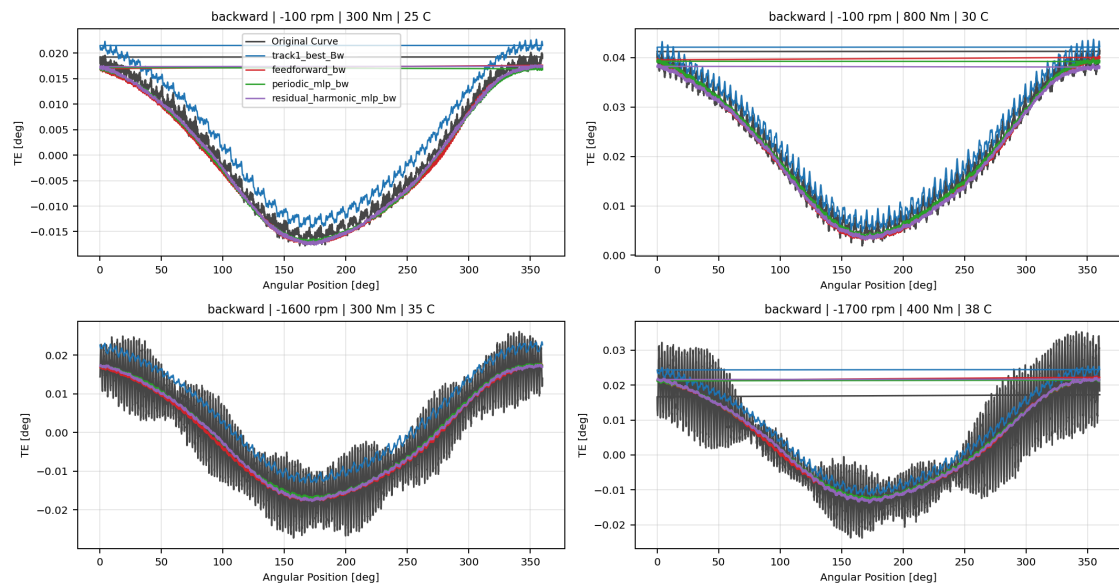


Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: `rcim_model_bank_reproduction_best_bw` , `feedforward_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` .

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

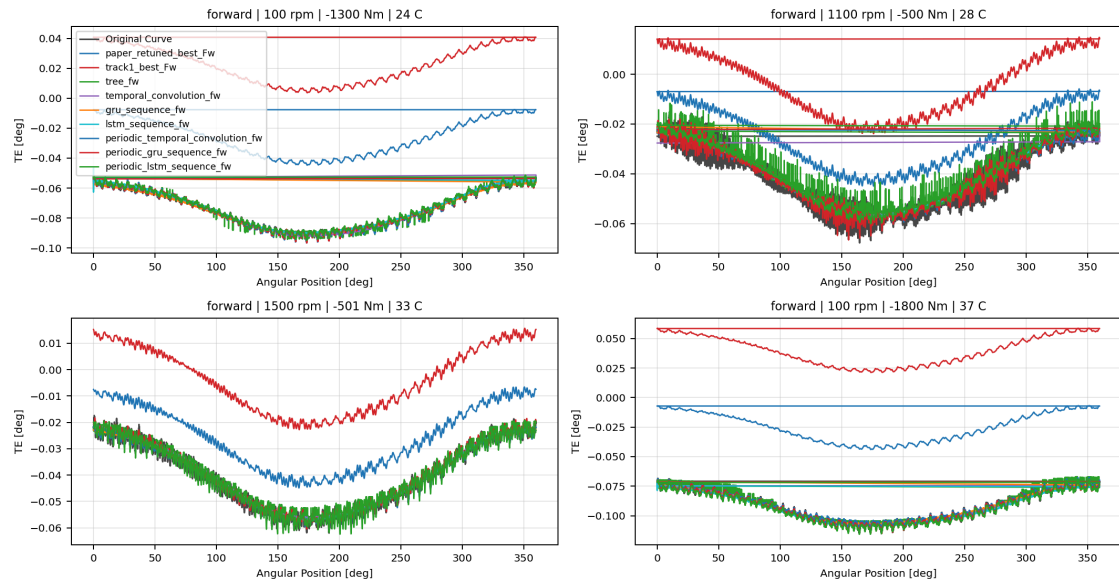


Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `temporal_convolution_fw`, `gru_sequence_fw`, `lstm_sequence_fw`, `periodic_temporal_convolution_fw`, `periodic_gru_sequence_fw`, `periodic_lstm_sequence_fw`.

Forward Reference Tree And Wave 2.1 Overlay

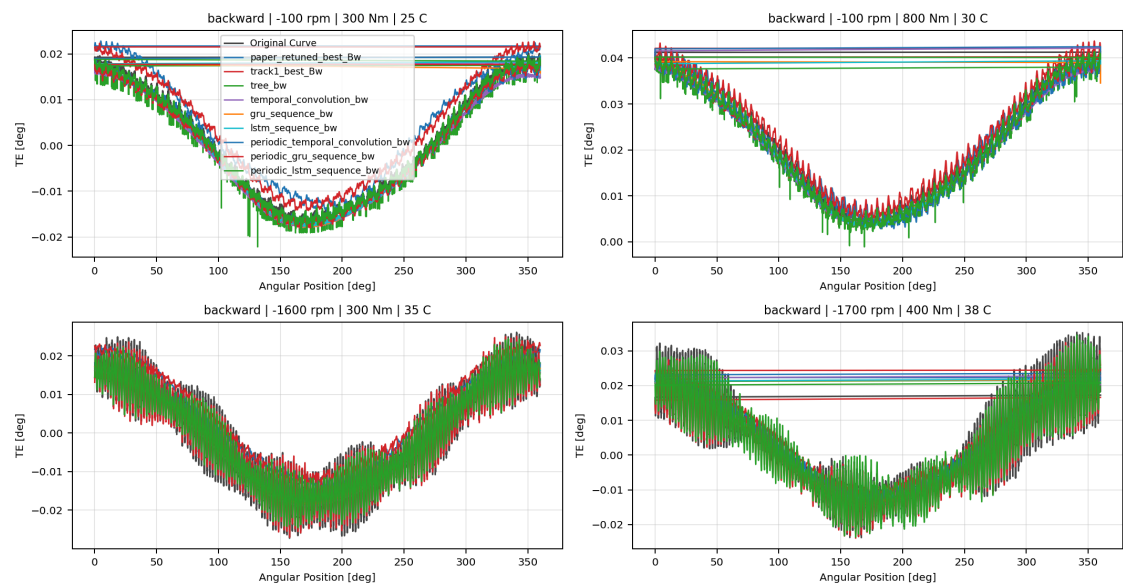


Forward Reference Tree And Wave 2.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Reference Tree And Wave 2.1 Overlay

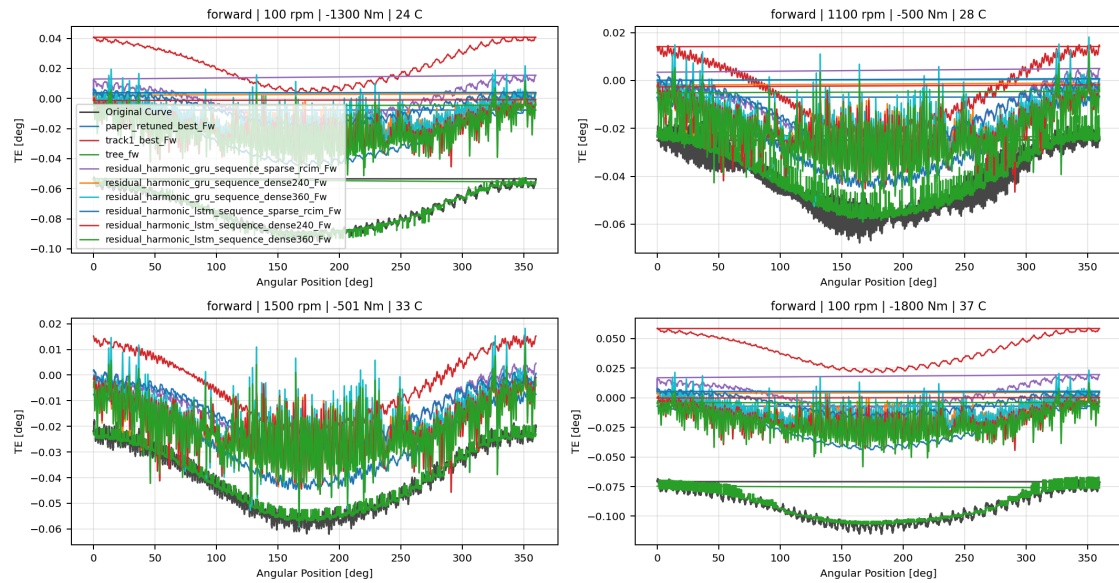


Backward Reference Tree And Wave 2.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `residual_harmonic_gru_sequence_sparse_rcim_Fw`, `residual_harmonic_gru_sequence_dense240_Fw`, `residual_harmonic_gru_sequence_dense360_Fw`, `residual_harmonic_lstm_sequence_sparse_rcim_Fw`, `residual_harmonic_lstm_sequence_dense240_Fw`, `residual_harmonic_lstm_sequence_dense360_Fw`.

Forward Reference Tree And Wave 2.3 Overlay

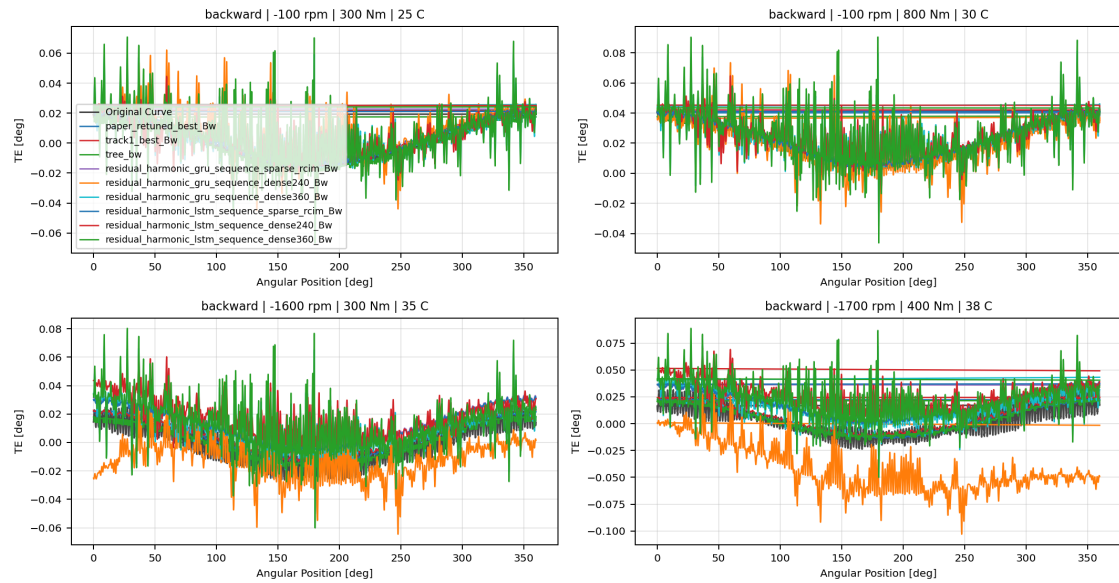


Forward Reference Tree And Wave 2.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `residual_harmonic_gru_sequence_sparse_rcim_Bw` , `residual_harmonic_gru_sequence_dense240_Bw` , `residual_harmonic_gru_sequence_dense360_Bw` , `residual_harmonic_lstm_sequence_sparse_rcim_Bw` , `residual_harmonic_lstm_sequence_dense240_Bw` , `residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Reference Tree And Wave 2.3 Overlay

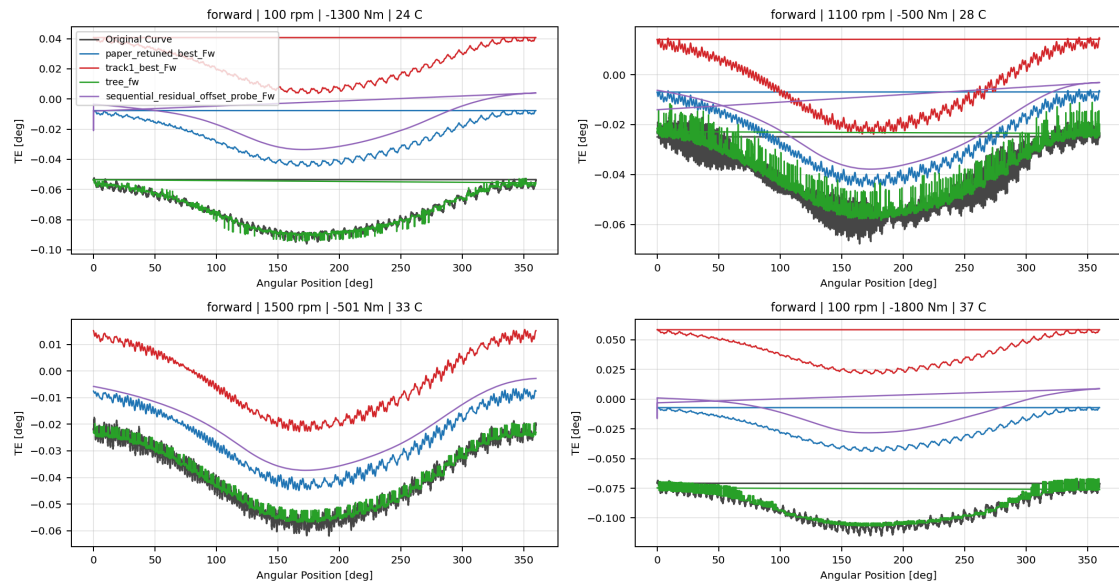


Backward Reference Tree And Wave 2.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `wave3_1_sequential_residual_offset_probe_fw`.

Forward Reference Tree And Wave 3.1 Overlay

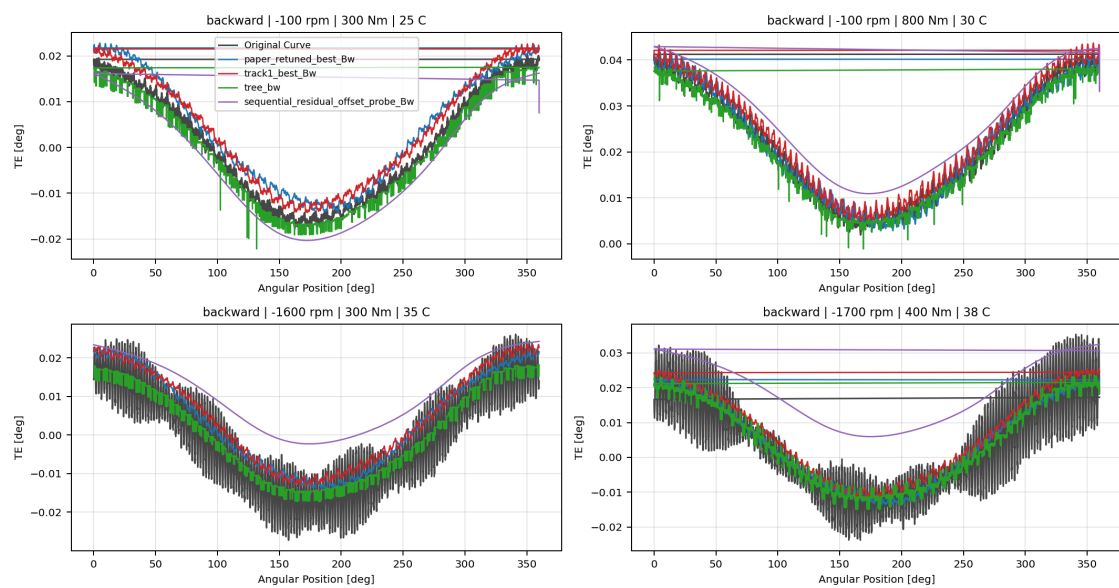


Forward Reference Tree And Wave 3.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Bw`, `rcim_model_bank_reproduction_best_bw`, `tree_bw`, `wave3_1_sequential_residual_offset_probe_bw`.

Backward Reference Tree And Wave 3.1 Overlay

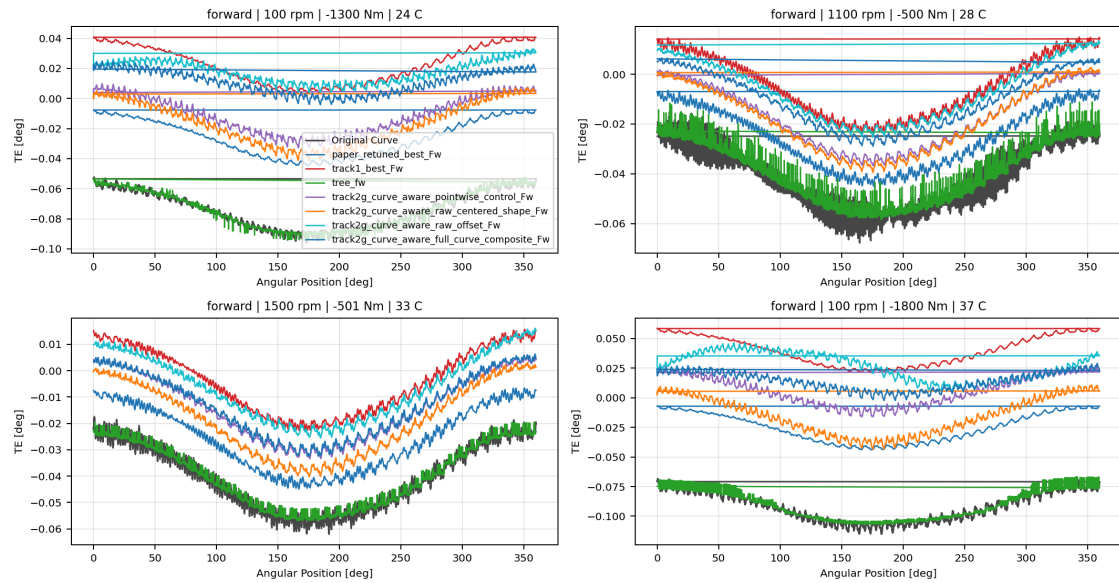


Backward Reference Tree And Wave 3.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` ,
`wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` ,
`wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Reference Tree And Wave 3.3 Overlay

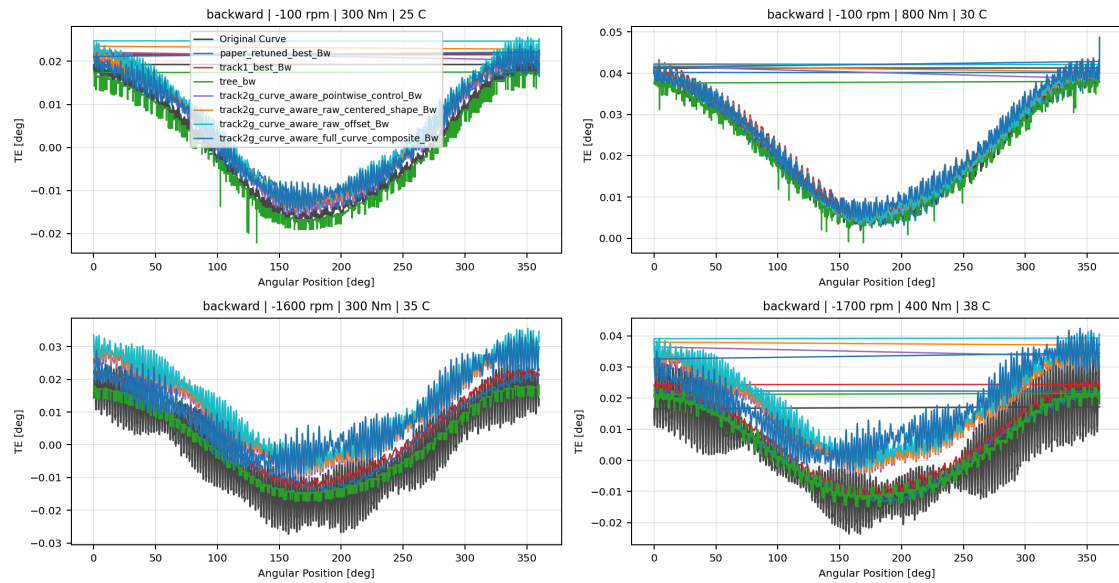


Forward Reference Tree And Wave 3.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_Bw`, `rcim_model_bank_reproduction_best_bw`, `tree_bw`, `wave3_3_curve_aware_pointwise_control_bw`, `wave3_3_raw_centered_shape_curve_aware_bw`, `wave3_3_raw_offset_curve_aware_bw`, `wave3_3_full_curve_composite_bw`.

Backward Reference Tree And Wave 3.3 Overlay

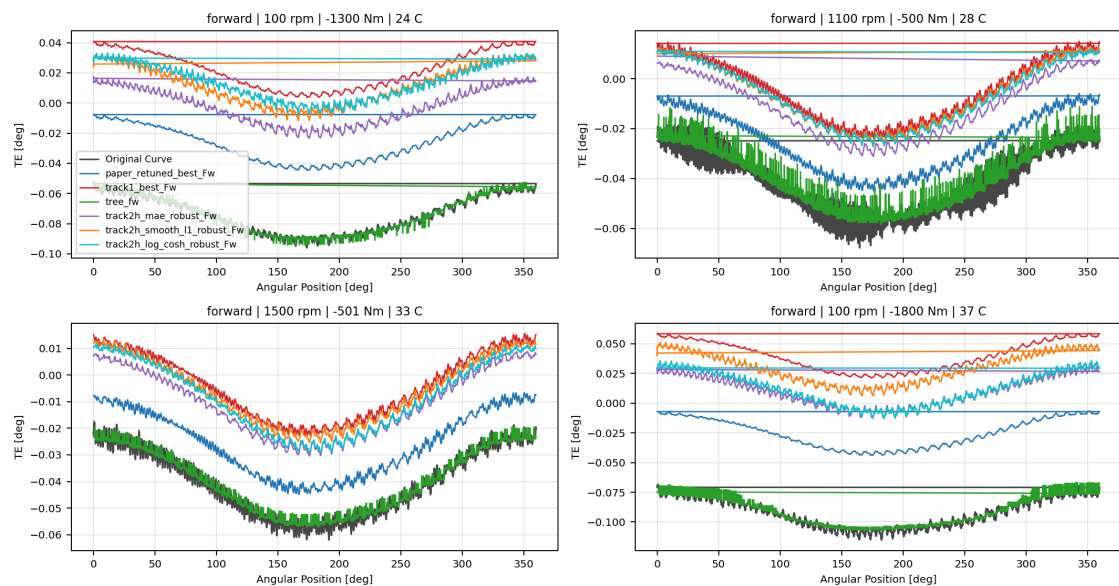


Backward Reference Tree And Wave 3.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `wave4_1_mae_robust_loss_fw`, `wave4_1_smooth_l1_robust_loss_fw`, `wave4_1_log_cosh_robust_loss_fw`.

Forward Reference Tree And Wave 4 series Overlay

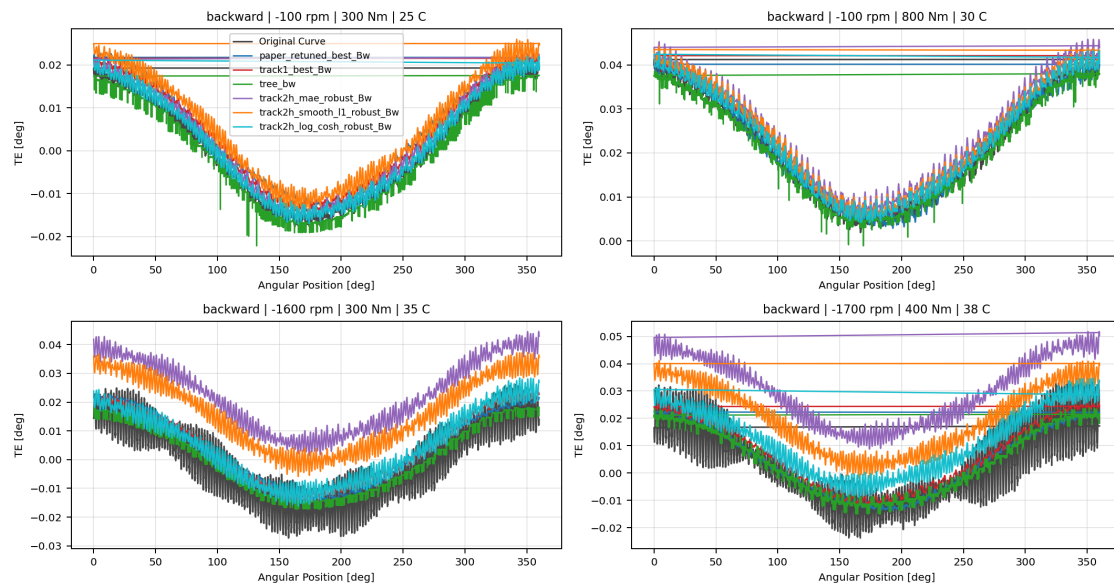


Forward Reference Tree And Wave 4 series Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Bw`, `rcim_model_bank_reproduction_best_bw`, `tree_bw`, `wave4_1_mae_robust_loss_bw`, `wave4_1_smooth_l1_robust_loss_bw`, `wave4_1_log_cosh_robust_loss_bw`.

Backward Reference Tree And Wave 4 series Overlay



Backward Reference Tree And Wave 4 series Overlay TE Curve Verification Pipeline comparison

Output Artifacts

- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-02-14-44-11__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-02-14-44-11__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-02-14-44-11__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\multi_model_curve_comparison_report\[2026-07-02]\track2_multi_model_curve_comparison_report.md` .